

Inflation since COVID: Demand or Supply

A. Cerrato and G. Gitti (2022)

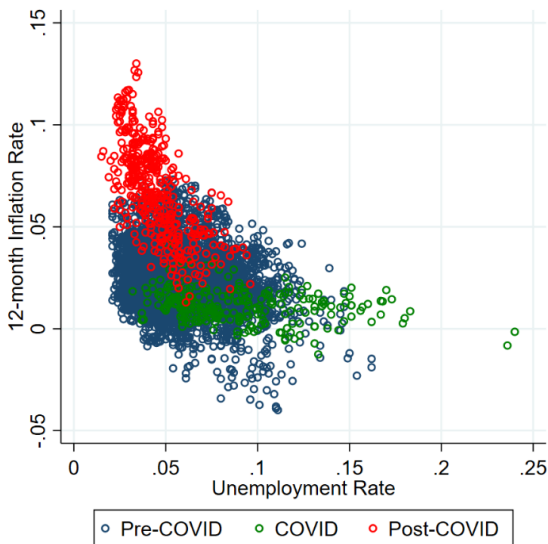
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Motivation: Phillips correlation and COVID

Figure 1: The Phillips Correlation Across US Cities



- Goal: Estimate slope of Phillips curve before/during/after COVID
- Strategy: Exploit heterogeneity across MSA-level inflation and unemployment rates
- 2-region NK-model $\implies$ aggregate PC slope = regional PC slope
- Bartik instruments to isolate demand-driven fluctuations in MSA unemployment rates
- Result: 1/4 of post-COVID inflation driven by demand

Outline

- ① Model
- ② Estimation
- ③ Results

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- New Keynesian model with two regions:
 - Home H , measure ζ
 - Foreign F , measure $1 - \zeta$
- Regional representative household
- Common monetary authority
- Three-sector supply chain

Representative H household

- Maximizes:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_{H,t}, N_{H,t})$$

- GHH preferences:

$$u(C_{H,t}, N_{H,t}) = \frac{\left(C_{H,t} - \chi_{H,t} \frac{N_{H,t}^{1+1/\phi}}{1+1/\phi} \right)^{1-1/\sigma}}{1 - 1/\sigma}$$

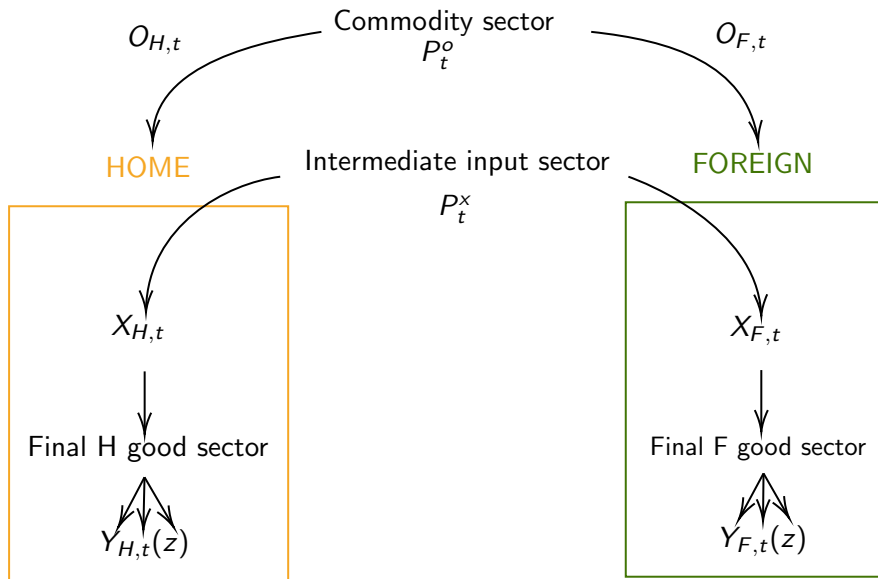
- Final consumption good is a composite:

$$C_{H,t} = \left[\int_0^1 C_{H,t}(z)^{\frac{\theta-1}{\theta}} dz \right]^{\frac{\theta}{\theta-1}}$$

- Budget constraint:

$$\int_0^1 C_{H,t}(z) P_{H,t}(z) dz + \mathbb{E}_t[M_{H,t,t+1} B_{H,t+1}] \leq B_{H,t} + W_{H,t} N_{H,t} + \int_0^1 \Pi_{H,t}(z) dz$$

Supply-side structure



- International market
- Inverse supply curve:

$$P_t^o = c_t^o O_t$$

- Supplied to intermediate-input sectors

Intermediate-Input sector

- Perfect competition, tradable
- Local firms
- Production function:

$$X_{H,t} = A_{H,t}^x (N_{H,t}^x)^\gamma O_{H,t}^{1-\gamma}$$

- Static optimization problem:

$$\max_{O_{H,t}, N_{H,t}^x} P_t^x X_{H,t} - W_{H,t} N_{H,t}^x - P_t^o O_{H,t}$$

- Monopolistic competition, local firms
- Calvo pricing, $\Pr(\text{update price}) = a$
- Production function:

$$Y_{H,t}(z) = A_{H,t}^y [X_{H,t}(z)]^{1-\alpha} [N_{H,t}^y(z)]^\alpha$$

- Dynamic optimization problem:

$$\mathbb{E}_t \sum_{k=0}^{\infty} M_{H,t,t+k} [P_{H,t+k}(z) Y_{H,t+k}(z) - W_{H,t+k} N_{H,t+k}(z) - P_{t+k}^x X_{t+k}]$$

- Demand for differentiated goods:

$$Y_{H,t}(z) = Y_{H,t} \left(\frac{P_{H,t}(z)}{P_{H,t}} \right)^{-\theta}$$

- Common monetary policy across regions
- Taylor rule:

$$r^n = \phi_\pi(\pi_t - \bar{\pi}_t) - \phi_u(\hat{u}_t - \bar{u}_t) + \varepsilon_{r,t}$$

- Economy-wide inflation:

$$\pi_t = \zeta \pi_{H,t} + (1 - \zeta) \pi_{F,t}$$

Regional vs aggregate Phillips curves

- Regional Phillips curve:

$$\pi_{H,t} = \beta \mathbb{E}_t \pi_{H,t+1} + \kappa \hat{n}_{H,t} + \lambda(1 - \alpha) \hat{p}_{H,t}^x + \lambda \alpha \hat{\chi}_{H,t} - \lambda \hat{a}_{H,t}^y$$

- Cost-push shocks:
 - $\hat{p}_{H,t}^x = \frac{\hat{P}_t^x}{P_{H,t}}$: SS deviation of regional price of intermediate input
 - $\hat{\chi}_{H,t}$: local shocks to households' disutility of labour
 - $\hat{a}_{H,t}^y$: local shocks to final-goods sector productivity
- Slope of regional Phillips curve:

$$\kappa = \frac{(1 - a\beta)(1 - a)}{a}$$

Derivations

Aggregate Phillips curve

- Given $\pi_t = \zeta\pi_{H,t} + (1 - \zeta)\pi_{F,t}$, aggregate Phillips curve is:

$$\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa\hat{n}_t + \lambda(1 - \alpha)\hat{p}_t^x + \lambda\alpha\hat{\chi}_t - \lambda\hat{a}_t^y$$

- κ unchanged by aggregation:

$$\kappa = \frac{(1 - a\beta)(1 - a)}{a}$$

Derivations

- Solve regional Phillips curve forward:

$$\pi_{H,t} = \mathbb{E}_t \pi_{t+\infty} - \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \kappa \tilde{u}_{H,t+j} + \sum_{j=0}^{\infty} \beta^j \lambda \left[(1-\alpha) \hat{p}_{H,t+j}^x + \alpha \hat{\chi}_{H,t+j} - \hat{a}_{H,t+j}^y \right]$$

- Assuming AR(1) processes gives

$$\pi_{H,t} = \mathbb{E}_t \pi_{t+\infty} - \psi \tilde{u}_{H,t} + \delta \hat{p}_{H,t}^x - \eta \hat{a}_{H,t}^y + \omega_{H,t}$$

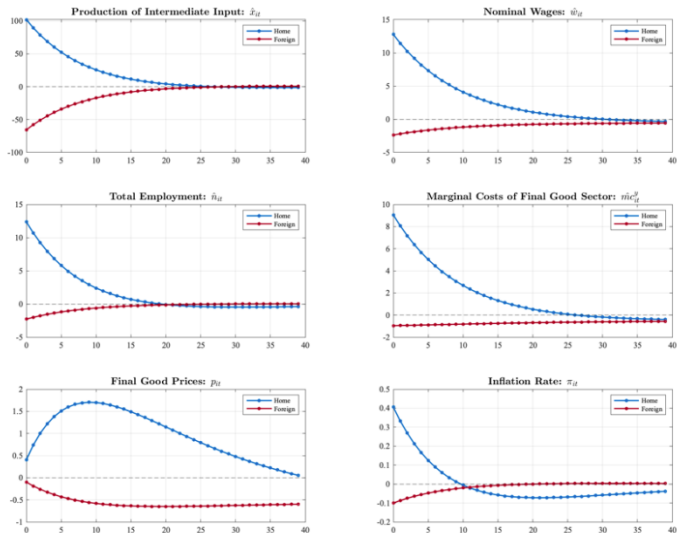
- Key variable here is ψ :

$$\psi = \frac{\kappa}{1 - \beta \rho_u}$$

Derivations

IRFs and identification strategy

Figure D.1: Impulse Response Functions to Intermediate Sector Productivity Shock



Outline

① Model

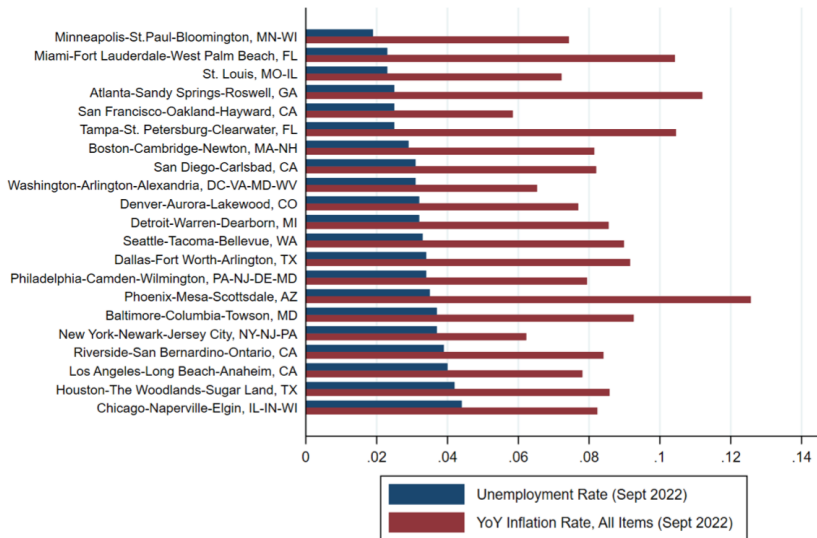
② Estimation

③ Results

- 21 US MSAs, January 1990 - September 2022
- Consumer Price Index:
 - From BLS, monthly or bimonthly
- Labour market:
 - From BLS Local Area Unemployment Statistics
 - Labour force, employment, unemployment, unemployment rate
- Shift-share instrument:
 - Shift: CPS monthly employment by industry
 - Share: MSA-level industry employment shares from 1990 Census

Unemployment and Inflation across MSAs

Figure 2: Unemployment and Inflation Across US Cities



- Recall regional Phillips curve:

$$\pi_{H,t} = \mathbb{E}_t \pi_{t+\infty} - \psi \tilde{u}_{H,t} + \delta \hat{p}_{H,t}^x - \eta \hat{a}_{H,t}^y + \omega_{H,t}$$

- Estimate following empirical model:

$$\pi_{i,t} = \alpha_i + \gamma_t - \psi u_{i,t} + \delta \hat{p}_{i,t}^x - \eta z_{i,t}^y + \omega_{i,t}$$

Instruments

- Shift-share instrument for $u_{i,t}$:

$$z_{i,t}^x = \underbrace{\sum_{j=1}^{N^x} \frac{E_{i,j,1990}}{E_{i,1990}} \times \Delta_{3Y} \log E_{US,j,t}}_{\text{Tradable intermediate-input industries}}$$

- Shift-share instrument for $\hat{a}_{i,t}^y$:

$$z_{i,t}^y = \underbrace{\sum_{j=1}^{N^y} \frac{E_{i,j,1990}}{E_{i,1990}} \times \Delta_{3Y} \log E_{US,j,t}}_{\text{Non-tradable final-goods industries}}$$

- Instrument for $\hat{p}_{i,t}^x$:

$$\hat{p}_{i,t}^{x*} = \frac{\hat{P}_t^x}{P_{i,t-24}}$$

First stage coefficients

Table 1: First Stage Coefficients

	u_{it}			$\hat{p}_{it}^x$		
	(1) Pre-COVID	(2) COVID	(3) Post-COVID	(4) Pre-COVID	(5) COVID	(6) Post-COVID
z_{it}^x	-0.49*** (0.09)	-2.01*** (0.53)	-0.71*** (0.16)	-0.19 (0.12)	-0.05 (0.14)	-0.31 (0.18)
$\hat{p}_{it}^{x*}$	0.02 (0.02)	0.23** (0.07)	0.02* (0.01)	0.86*** (0.04)	0.55*** (0.06)	0.58*** (0.02)
z_{it}^y	-0.10 (0.16)	0.05 (0.13)	-0.48*** (0.10)	-0.20 (0.18)	0.09** (0.03)	-0.16 (0.15)
Observations	5211	252	399	5211	252	399
F-Statistic	25.11	11.80	9.40	273.69	78.27	544.94
MSA-Period FE	✓	✓	✓	✓	✓	✓
Year-Quarter FE	✓	✓	✓	✓	✓	✓

Outline

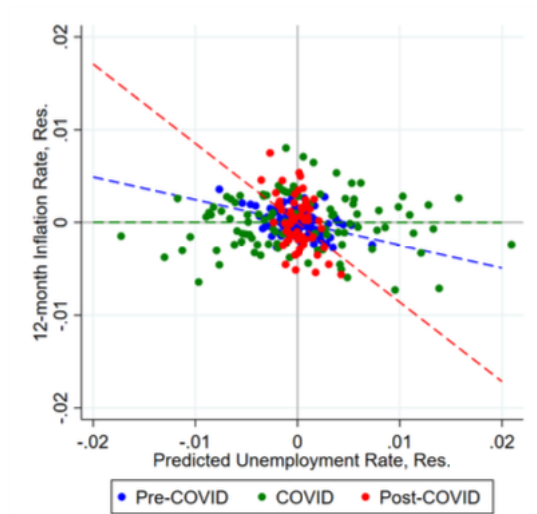
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Estimates for ψ

Table 2: Estimates of ψ from Equation (5)

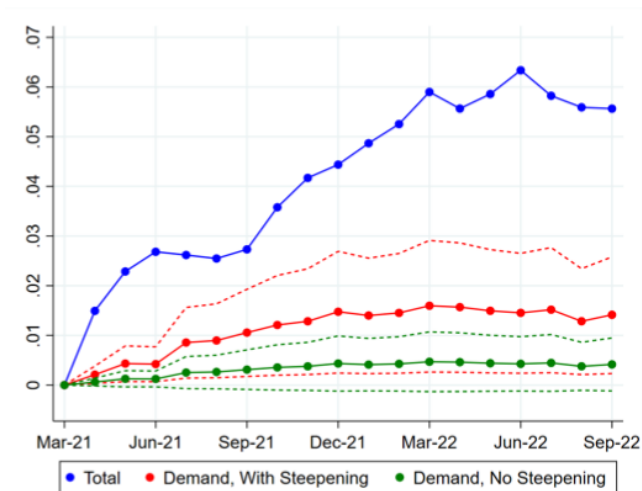
	(1) OLS	(2) OLS	(3) OLS	(4) IV
Pre-COVID				
u_{it}	-0.18*** (0.03)	-0.29*** (0.04)	-0.20*** (0.04)	-0.25 (0.15)
$\hat{p}_{it}^x$		0.09*** (0.02)	-0.06*** (0.02)	0.06** (0.03)
z_{it}^y				0.13 (0.09)
COVID				
u_{it}	-0.09*** (0.02)	-0.08*** (0.02)	-0.04 (0.02)	0.02 (0.07)
$\hat{p}_{it}^x$		0.02 (0.05)	-0.23** (0.08)	0.33*** (0.08)
z_{it}^y				0.01 (0.04)
Post-COVID				
u_{it}	-1.36*** (0.15)	-0.71*** (0.12)	0.10 (0.13)	-0.85** (0.34)
$\hat{p}_{it}^x$		0.24*** (0.03)	-0.08 (0.05)	0.20*** (0.04)
z_{it}^y				-0.14 (0.16)
Observations	5862	5862	5862	5862
MSA-Period FE	✓	✓	✓	✓
Year-Quarter FE			✓	✓

Figure 3: Estimates of ψ from column (4) of Table 2



The steepened Phillips curve

Figure 4: 12-month Inflation Rate, Change relative to March 2021



Contributions of goods and services

Table 4: IV Estimates of ψ from Equation (5) for broad CPI categories

	Goods			Services		
	(1) Pre-COVID	(2) COVID	(3) Post-COVID	(4) Pre-COVID	(5) COVID	(6) Post-COVID
u_{it}	-0.12 (0.15)	-0.21** (0.08)	-2.24*** (0.65)	-0.32 (0.23)	0.10 (0.06)	-0.35 (0.36)
$\hat{p}_{it}^x$	0.24*** (0.07)	0.65** (0.27)	0.64*** (0.09)	0.05** (0.02)	0.08 (0.07)	0.09** (0.04)
z_{it}^y	0.08 (0.07)	-0.07 (0.10)	-0.57* (0.31)	0.12 (0.15)	0.04 (0.04)	0.01 (0.19)
Observations	5211	252	399	5211	252	399
MSA-Period FE	✓	✓	✓	✓	✓	✓
Year-Quarter FE	✓	✓	✓	✓	✓	✓

- Q: What was the slope of the PC before, during and after COVID?
- A: Flattening of PC during COVID, steepening of PC after COVID
 - 1/4 of post-COVID inflation due to smaller unemployment gap
 - Slope of PC increased more in early phase of post-COVID
 - Slope of PC impacted more by goods than services

A1: Phillips curves derivation



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A2: Phillips curves' slopes derivation



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